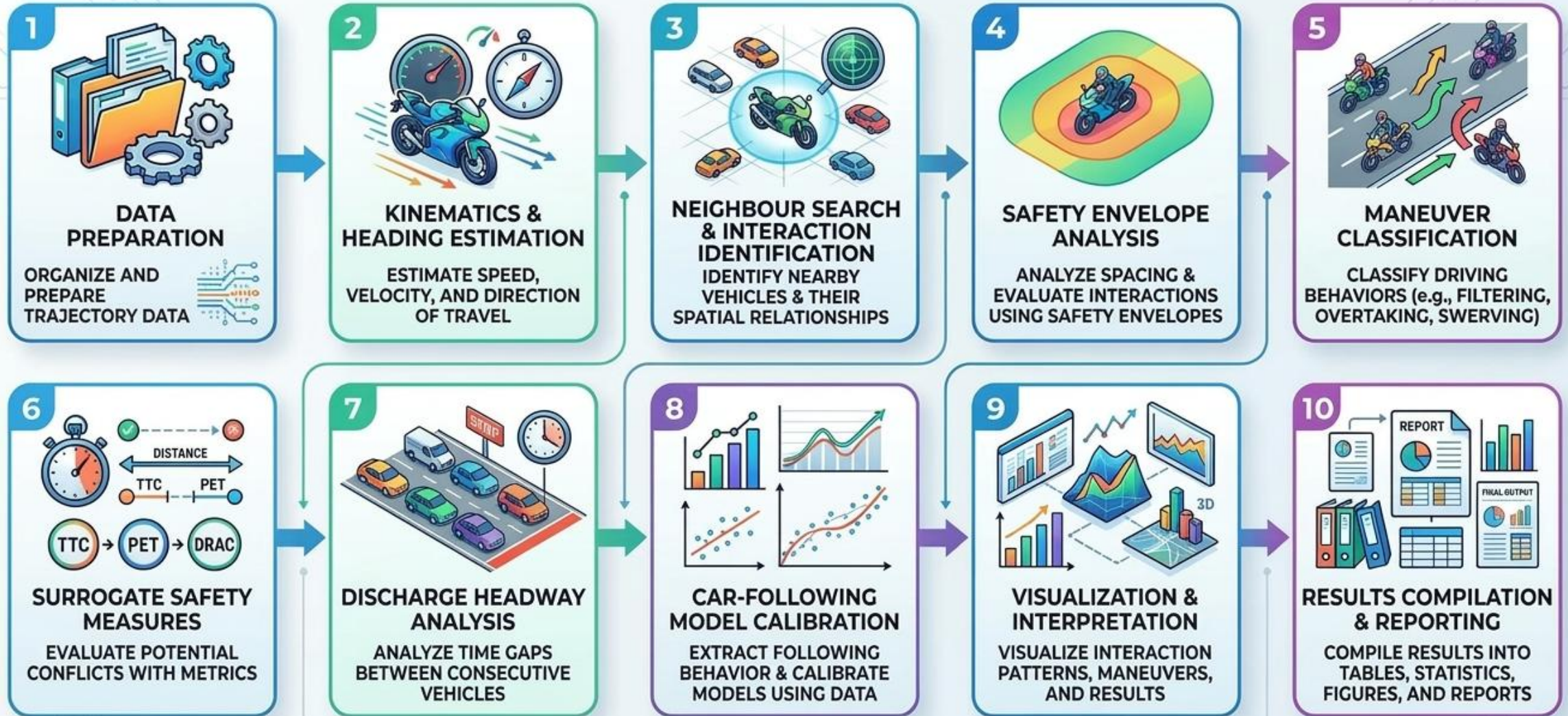


Framework for Naturalist Motorcycle Driving Behavior Analysis at Signalized Intersection Approaches

Prepared by: Saurav Barua

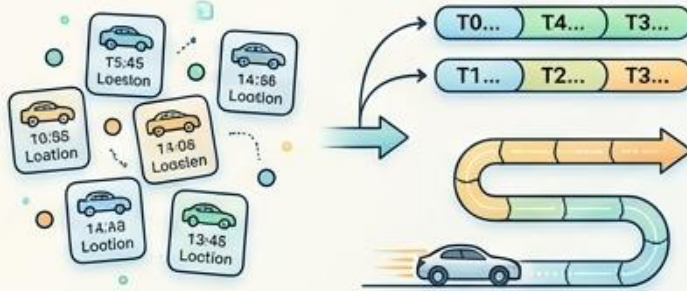
FRAMEWORK FOR NATURALIST MOTORCYCLE DRIVING BEHAVIOR ANALYSIS



KINEMATICS & HEADING ESTIMATION

PROCESSING VEHICLE TRAJECTORY DATA

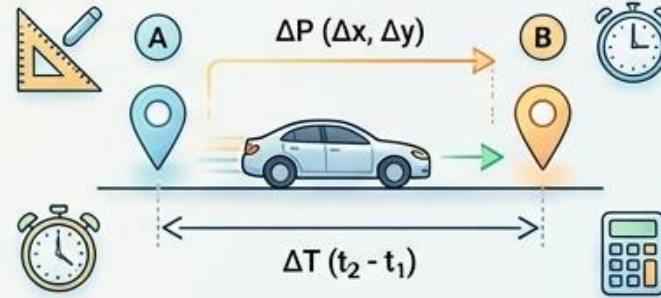
1 Trajectory Organization



Arrange the trajectory observations chronologically for each vehicle.

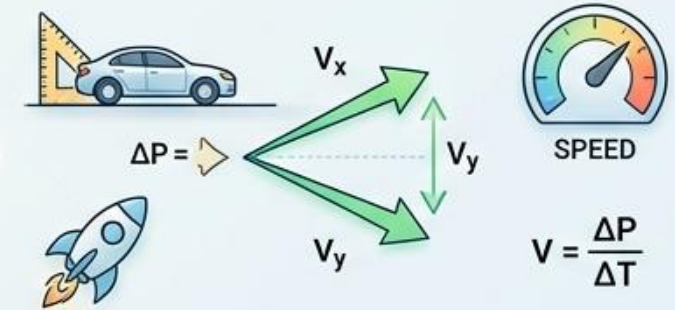
INPUT: Raw Data points → OUTPUT: Sorted Sequences (e.g., T1, T2, T3...)

2 Position and Time Differences



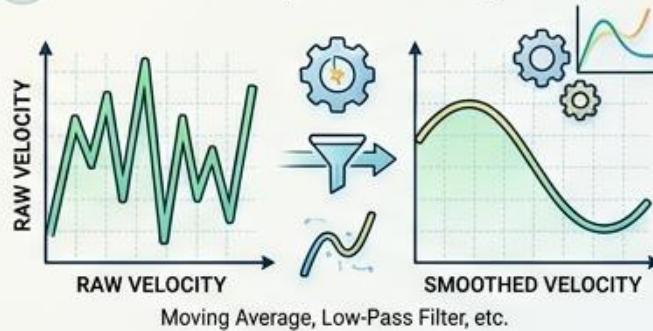
Calculate changes in position and time between consecutive observations.

3 Velocity Estimation



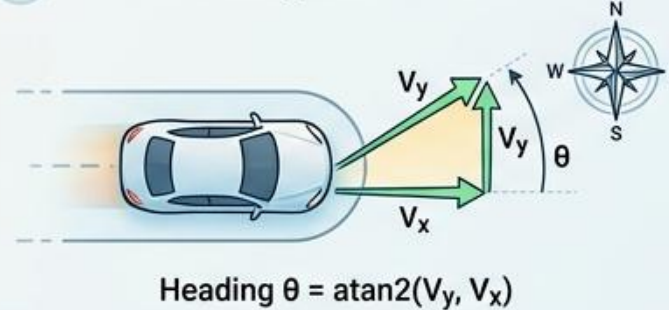
Estimate the velocity components (V_x, V_y) from the position and time differences.

4 Velocity Smoothing



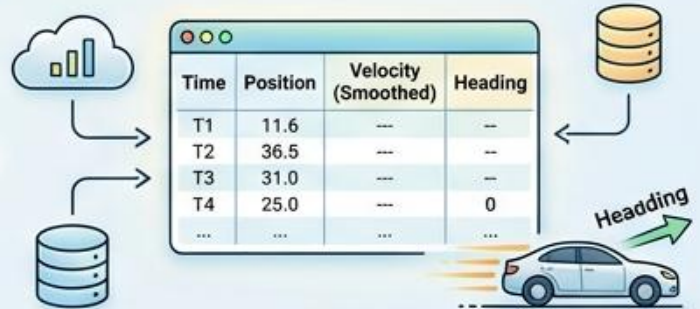
Smooth the velocity components to reduce fluctuations and tracking noise.

5 Heading Estimation



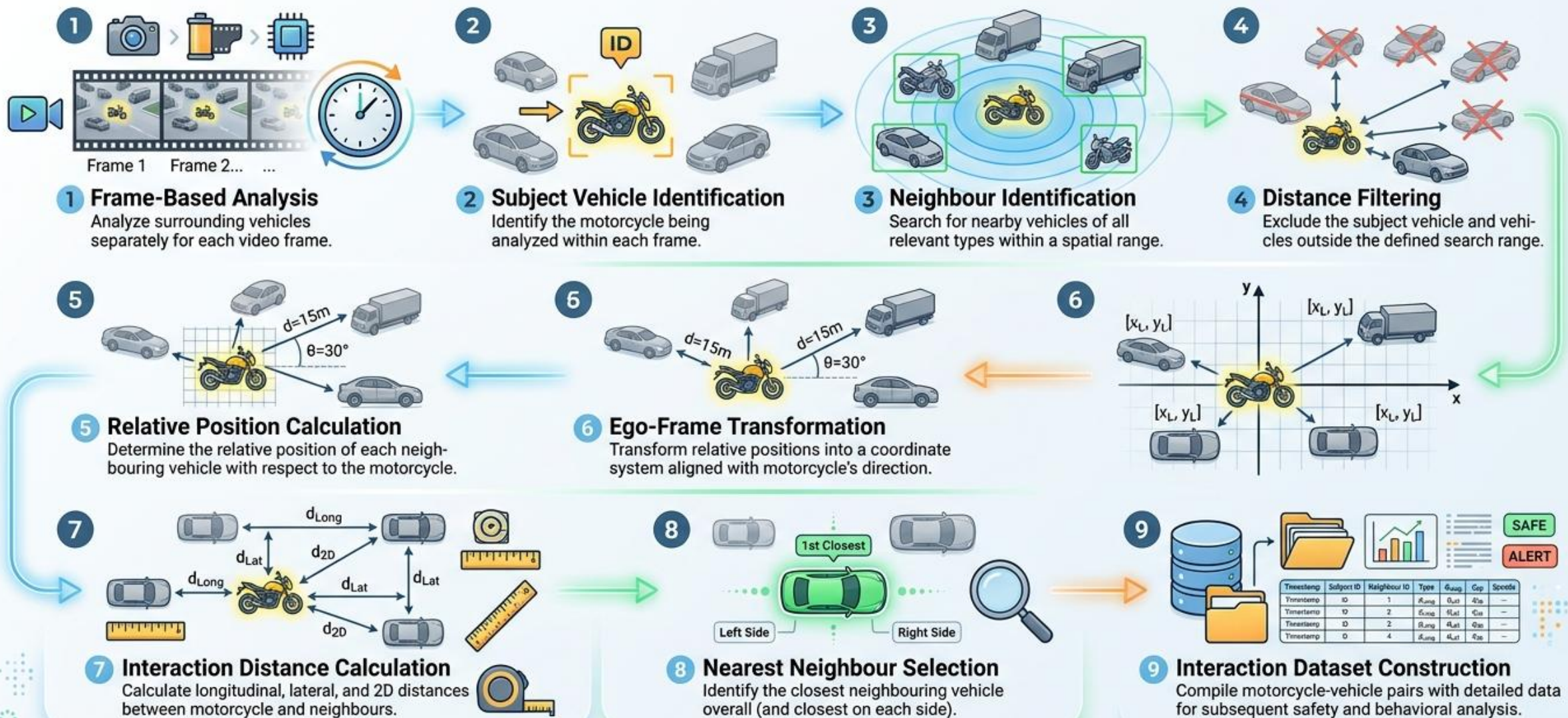
Determine the vehicle's heading from the smoothed velocity components using the two-argument arctangent.

6 Kinematic Dataset Preparation



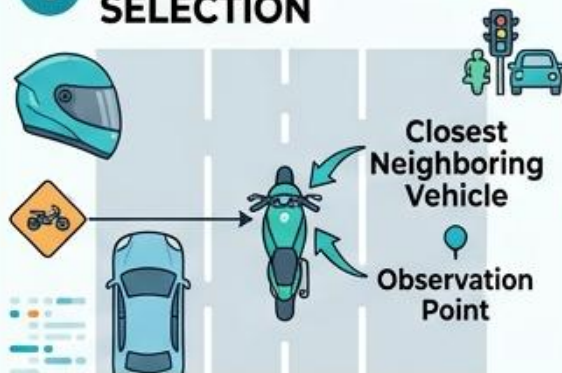
Add the estimated velocity components and heading to the trajectory data for subsequent interaction analysis.

NEIGHBOUR SEARCH AND INTERACTION IDENTIFICATION



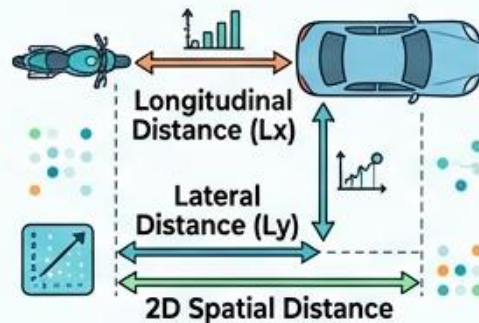
SAFETY ENVELOPE ANALYSIS FRAMEWORK

1 PRIMARY INTERACTION SELECTION



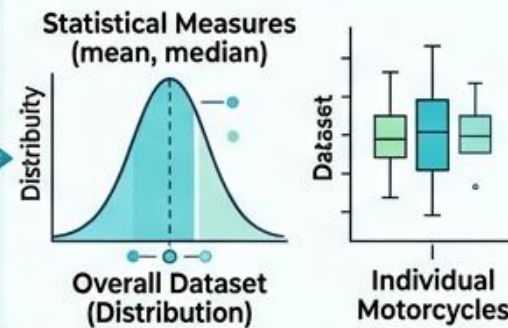
Select closest neighbor and observation

2 INTERACTION SPACING ANALYSIS



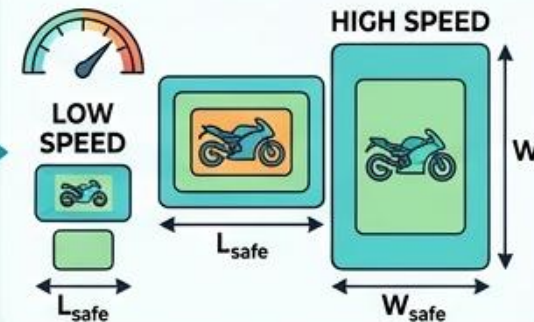
Calculate distances

3 EMPIRICAL ENVELOPE CHARACTERIZATION



Determine typical interaction spacing

4 SPEED-DEPENDENT ENVELOPE DEFINITION



Define longitudinal & lateral dimensions based on Speed & Size

5 ENVELOPE INDEX CALCULATION

$$INDEX = \frac{\text{Observed Spacing}}{\text{Envelope Dimension}}$$

$$\text{Lateral} = \frac{L_y}{L_{safe}}$$

$$\text{Longitudinal } x = \frac{L_x}{W_{safe}}$$

Normalize spacings

6 ENVELOPE STATUS CLASSIFICATION



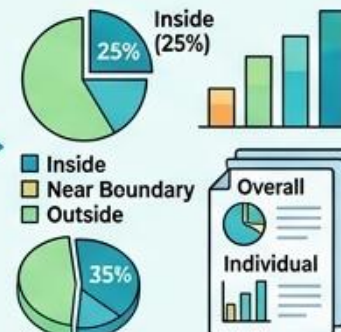
Classify each interaction

7 INTERACTION-LEVEL ASSESSMENT



Generate interaction-level dataset

8 STATISTICAL SUMMARY



Summarize distribution & counts

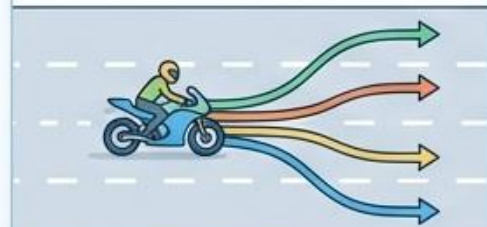
9 SAFETY ENVELOPE INTERPRETATION



Assess frequency & degree of interactions. Define parameters & validate.

MOTORCYCLE MANEUVER CLASSIFICATION PROCESS

1 MOTORCYCLE TRAJECTORY SELECTION



Select and analyze each motorcycle trajectory independently.

2 KINEMATIC FEATURE EXTRACTION



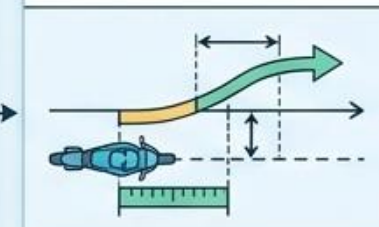
Obtain speed, lateral acceleration, position, and duration from trajectories.

3 REFERENCE DIRECTION ESTIMATION



Determine the motorcycle's overall direction of travel from its trajectory.

4 LATERAL DISPLACEMENT ESTIMATION



Calculate lateral movement relative to the overall travel direction.

5 LATERAL DISPLACEMENT ESTIMATION



Examine surrounding vehicles to identify closely confined movement conditions.

5 NEIGHBOUR-GAP ASSESSMENT



Examine surrounding vehicles to identify closely confined movement conditions.

7 BEHAVIOURED INDICATOR CALCULATION



8 OVERTAKING & LANE-CHANGE DIFFERENTIATION



Use position and speed of the vehicle ahead to distinguish overtaking from general lane-changing.

9 MANEUVER DATASET CONSTRUCTION



Record the classified maneuver together with supporting behavioral indicators.

10 CLASSIFICATION VALIDATION



Treat classifications as heuristic results and validate them against manually reviewed video observations.

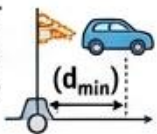
SURROGATE SAFETY MEASURES (SSMs) CALCULATION PROCESS

Data Selection → Approach Analysis → SSMs Estimation → Validity & Summary → Conflict Assessment

1 PRIMARY INTERACTION SELECTION



Select the **CLOSEST** neighboring vehicle for each motorcycle and observation.

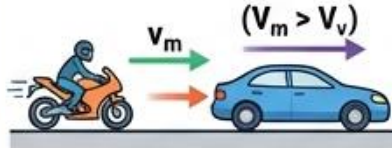


2 SAME-PATH INTERACTION FILTERING



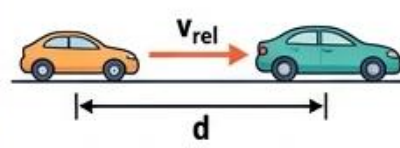
Retain interactions where vehicles are sufficiently close **LATERALLY** to represent the same path.

3 APPROACHING VEHICLE IDENTIFICATION



Identify cases where the neighboring vehicle is **AHEAD** and the motorcycle is **CLOSING** based on relative speed.

4 TIME-TO-COLLISION (TTC)

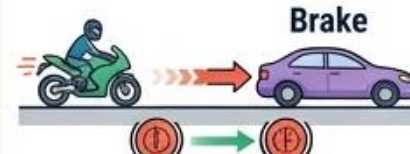


Estimate time required for motorcycle to reach vehicle ahead.

$$TTC = \frac{d}{v_{rel}}$$

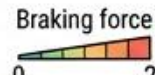


5 DECELERATION RATE TO AVOID COLLISION (DRAC)

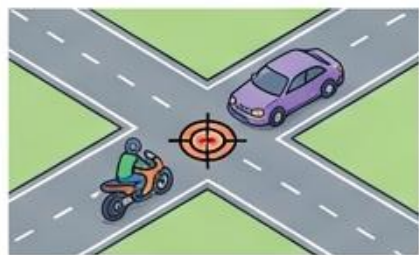


Estimate deceleration required to avoid collision.

$$DRAC = \frac{(v_{rel})^2}{2 \cdot d}$$

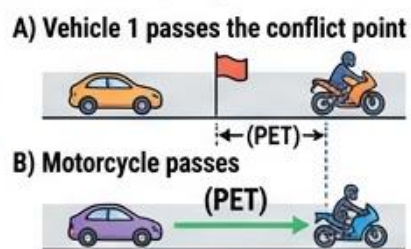


6 CLOSEST-APPROACH IDENTIFICATION FOR PET



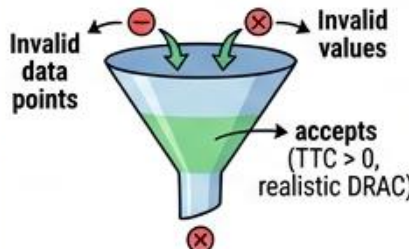
Examine each motorcycle-vehicle pair & identify the point of **CLOSEST APPROACH**.

7 POST-ENCROACHMENT TIME (PET)



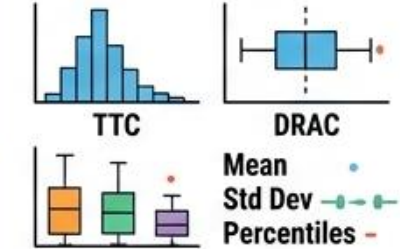
Estimate time difference associated with passage through potential conflict location.

8 VALIDITY FILTERING



Remove values outside predefined physically meaningful ranges for TTC, DRAC, and PET.

9 STATISTICAL SUMMARY



Calculate descriptive statistics (sample size, mean, std. deviation, percentiles) for each measure.

10 CONFLICT ASSESSMENT

Use TTC, PET, and DRAC to characterize **DIFFERENT ASPECTS** of potential motorcycle-vehicle conflicts.



Comprehensive Risk Profile

DISCHARGE HEADWAY ANALYSIS PROCESS

1 VEHICLE IDENTIFICATION



Identify the first observed record for each vehicle.

2 VEHICLE SELECTION



Select the relevant vehicle types for analysis.

3 GATE GROUPING



Group vehicles according to their common entry/reference gate.

4 TIME ORDERING



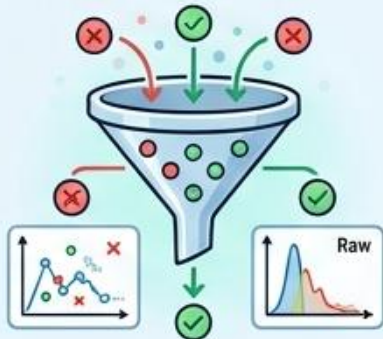
Arrange vehicles by their arrival or crossing time.

5 HEADWAY CALCULATION



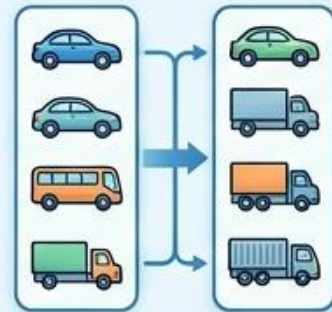
Calculate the time difference between consecutive vehicles.

6 DATA FILTERING



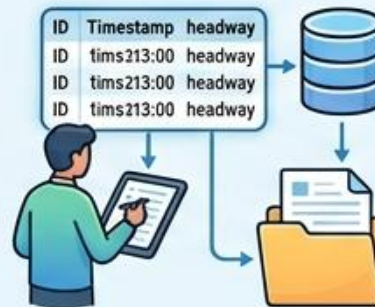
Remove non-positive and excessively large headway values.

7 VEHICLE CLASSIFICATION



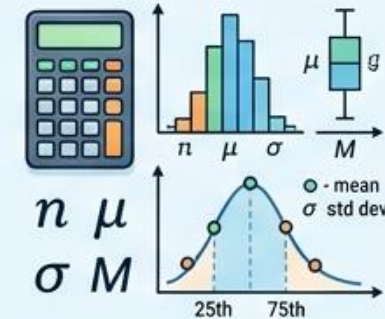
Classify headways by consecutive vehicle-type combinations.

8 OBSERVATION RECORDING



Record each valid headway observation for further analysis.

9 STATISTICAL ANALYSIS



Calculate sample size, mean, standard deviation, median, and selected percentiles for each vehicle combination.

10 INTERPRETATION

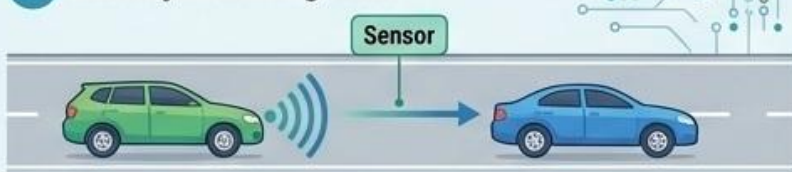


Interpret the measure as an arrival/discharge headway at the reference location, rather than a general time gap along the trajectory.

CAR-FOLLOWING MODEL CALIBRATION PROCESS

A. Following-Sample Extraction

A1. Identify Following Pairs



Select the closest preceding vehicle for each subject vehicle and observation.

A2. Filter Valid Interactions

- Vehicle Ahead
- Sufficiently Close
- Same Travel Path

Retain vehicles that are ahead, sufficiently close, and approximately in the same travel path.

A3. Match Observations



Match each following interaction with the subject vehicle's acceleration and calculate relative speed.

A4. Remove Invalid Data



Exclude missing, unrealistic, or physically implausible acceleration, spacing, speed, and relative-speed observations.

A5. Retain Following Samples



Use the remaining observations as representative car-following situations.

B. Model Formulation

B1. Define Model Parameters

- Desired Speed
- Desired Time Gap
- Acceleration
- Comfortable Deceleration
- Minimum Spacing

Specify desired speed, desired time gap, acceleration, comfortable deceleration, and minimum spacing parameters.

B2. Calculate Desired Spacing



Estimate desired spacing using speed, relative speed, and model parameters.

B3. Predict Acceleration



Calculate acceleration from the vehicle's speed, relative speed, and spacing.

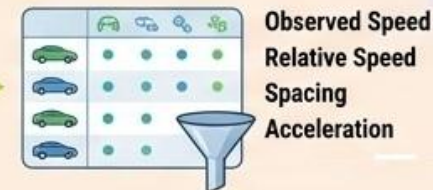
B4. Apply Physical Limits



Constrain predicted acceleration within a realistic range.

C. Parameter Calibration and Evaluation

C1. Prepare Calibration Data



Use observed speed, relative speed, spacing, and acceleration, with sampling reduced if computationally necessary.

C2. Calibrate Parameters



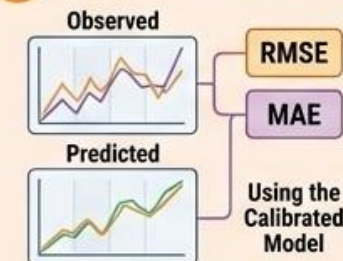
Minimize the difference between observed and predicted acceleration within predefined parameter ranges.

C3. Select Optimal Parameters



PARAMETER & PARAMETERS
Identify the parameter set producing the lowest calibration error.

C4. Evaluate Model Performance



Calculate RMSE, MAE, and using the calibrated model.

C5. Report Results



Provide calibrated parameters and observation-level observed versus predicted acceleration for further analysis.